

Price Discovery Under Market Lock: Black-Scholes Validity and Option Behavior During Queue Regimes in the Tehran Stock Exchange

Abstract

We study whether the Black-Scholes (BS) option-pricing framework holds for options on Ahrom (اهرم), a daily-rebalancing leveraged exchange-traded fund traded on the Tehran Stock Exchange (TSE), and we examine how those options behave when the underlying is locked in a price-limit queue. The TSE imposes a daily price-movement band—about $\pm 10\%$ in the earlier part of our sample, tightened to about $\pm 4\%$ from mid-2025—and when demand or supply is exhausted at a band edge the underlying becomes effectively untradeable while its options keep trading, creating a natural experiment in price discovery under a market lock. Using a 19-month panel of 450 option contracts (225 calls, 225 puts), the prevailing risk-free rate of roughly 30–40% per annum, and about 28 million reconstructed order-book snapshots, we organize the analysis along two axes. **Axis A** tests BS in the free (no-queue) regime; **Axis B** tracks option behavior as a function of queue age τ . All results are conditional on the liquid sub-universe (about 55.8% of contract-days within an instrument’s active life have zero volume). We report four results. First, once moneyness is measured correctly as K/S , near-term call implied volatility shows **no statistically significant monotonic skew** (HAC OLS slope -0.179 , $t = -1.41$, $p = 0.16$, $N = 1,744$); an apparent strong skew under the alternative S/K definition was an artifact. Second, BS **systematically underprices** options against an independent EWMA-volatility benchmark, with a median error of 221 rial and a mean of 452 rial (HAC $t = 18.67$, $p \approx 10^{-78}$, $N = 18,012$), and the underpricing now spans **both calls and puts**. Third, under the real risk-free rate the put-call parity basis is significantly **negative** (median -332 rial, mean -515 , HAC $t = -10.31$, $p \approx 10^{-25}$, $N = 6,972$): calls are cheap relative to puts, consistent with short-selling constraints that block the correcting arbitrage. Fourth, the parity-implied shadow price carries a statistically significant but economically negligible signal about the underlying’s next-day open (OLS slope 7.30×10^{-6} , $p \approx 0$, $N = 6,669$). Within queue episodes, implied volatility compresses with queue age (median IV falling from about 0.9 in the first minute to about 0.34 in the 5–30 minute range), and buy-queue quoting activity stays an order of magnitude below sell-queue activity. We attribute the underpricing to the leveraged fund’s volatility drag, and the negative parity basis to the high domestic interest rate combined with arbitrage frictions, and we conclude that BS is an imperfect descriptive model for this market even in its most favorable regime.

Keywords: Black-Scholes; implied volatility; put-call parity; price limits; trading halts; price discovery; leveraged ETF; risk-free rate; Tehran Stock Exchange; market microstructure; emerging markets.

1. Introduction

The Black-Scholes-Merton framework rests on a small set of structural assumptions: a continuously tradeable underlying that follows geometric Brownian motion with constant volatility, frictionless hedging, and no arbitrage. Decades of empirical work have documented that real option markets violate these assumptions in systematic ways. The bulk of this evidence comes from deep, continuously traded markets in developed economies. Far less is known about how the framework performs in a market whose microstructure deliberately interrupts continuous trading and whose institutional setting features very high nominal interest rates.

The Tehran Stock Exchange offers an unusually sharp setting in which to ask this question. Each security may move at most a fixed percentage from its prior-day reference price within a session—a band that was about $\pm 10\%$ in the earlier part of our sample and was tightened to about $\pm 4\%$ from roughly mid-2025. When buying interest is strong enough to clear the entire ask side at the ceiling, the market enters a **buy queue**: buyers stack at the ceiling price, no shares are offered, and the underlying becomes effectively frozen for the remainder of the session. A symmetric **sell queue** forms at the floor. Crucially, the options on a queued underlying continue to trade. During such an episode the option market becomes the only venue in which the market’s belief about the “true” underlying price can be expressed. This makes the TSE a laboratory for two distinct questions that we keep deliberately separate.

The first question is whether BS holds at all in this market under the conditions most favorable to it—when the underlying is free and continuously tradeable. If the model fails even here, the failure is attributable to the model, the asset, and the institutional setting, not to the lock. The second question is what happens to option behavior precisely when the lock is in force: does price discovery migrate to the options, and does the price the options imply for the locked underlying contain information?

Two features of the setting shape every result. The underlying, Ahrom, is a **daily-rebalancing leveraged ETF**: it resets its leverage each day, which makes its multi-day returns path-dependent and subject to *volatility drag*, so that a historical-volatility estimate understates the volatility relevant for pricing its options. And the domestic **risk-free rate is high**—roughly 30–40% per annum over the sample—so the present-value discount on a strike is large and materially affects both the BS price and the put-call parity relationship. Both features violate or strain the textbook BS assumptions independently of the market lock.

Our contribution is threefold. (i) We assemble and clean a large intraday order-book panel for a leveraged-ETF option complex on an emerging market that operates hard, time-varying price limits, documenting a structured illiquidity (about 55.8% zero-volume contract-days) that governs every downstream choice. (ii) We provide conditional, regime-stratified evidence on BS validity—IV skew, term structure, cross-contract dispersion, pricing error, and parity—in the free regime, using the real risk-free rate throughout. (iii) We introduce a queue-age clock τ and a parity-implied *shadow price* to characterize option dynamics under the lock, and we test whether that shadow price forecasts the underlying’s next-day open. Throughout, every reported result is stated conditionally on its regime, moneyness, maturity, and the liquidity gate.

The remainder of the paper is organized as follows. Section 2 situates the study in the literature. Section 3 describes the market structure and data. Section 4 sets out the methodology. Section 5 presents Axis A; Section 6 presents Axis B. Section 7 discusses

2. Related Literature

The study connects four strands of work, and is intended to be read against them.

Tests of Black-Scholes and the volatility smile. The canonical option-pricing model of Black and Scholes (1973) predicts a single volatility consistent across strikes and maturities. The empirical literature has instead documented implied-volatility smiles and skews, term-structure effects, and systematic pricing biases, motivating stochastic-volatility and jump-diffusion extensions. Our free-regime evidence belongs to this tradition, but with a cautionary twist: the apparent monotonic skew we first found dissolves once moneyness is defined correctly, underscoring how sensitive “stylized facts” can be to measurement conventions.

Trading halts, circuit breakers, and price limits. A large microstructure literature studies how mandatory trading interruptions affect price discovery, volatility, and information transmission. Kim and Rhee (1997) frame three competing hypotheses—volatility spillover, delayed price discovery, and trading interference—and find evidence consistent with all three on the Tokyo Stock Exchange. Price-limit queues on the TSE are a recurring, endogenous form of this interruption, and the continued trading of options during a lock lets us observe where discovery goes when the cash market is closed.

Options as a venue for price discovery. When a stock is halted but its options trade, the options can reveal the market’s view of the unobserved underlying price. Put-call parity provides the natural instrument: the combination $C - P + Ke^{-rt}$ recovers an implied (shadow) underlying value. We adopt this construction directly—now with the real r —and test its informativeness.

Leveraged and inverse ETFs. Work on daily-rebalancing leveraged funds shows that their returns are path-dependent and that volatility drag erodes compounded performance, complicating the pricing of options written on them (Avellaneda and Zhang, 2010). This literature provides the mechanism we invoke to explain why an independent historical-volatility benchmark systematically under-prices Ahrom options.

We do not attempt an exhaustive survey here; the reference list collects the foundational works in each of these areas. The point of the synthesis is that our setting sits at their intersection: a leveraged-ETF option complex, on an emerging market, under hard price limits and high interest rates, observed at intraday resolution.

3. Market Structure and Data

3.1 Instruments

The instrument family consists of one underlying and its two option classes. The underlying, Ahrom (اهرم), is a leveraged ETF that pays no dividend and is highly liquid. The call options trade under the family اختیاری اهرم and the puts under اختیاری اهرم. Option type is re from the data field rather than inferred from the symbol. Following standard practice for TSE equity options and the available contract specifications, options are treated as European; confirmation of exercise style against the exchange rulebook is listed among the open items in Section 8.

3.2 Data sources and coverage

Four datasets underpin the study.

The **underlying daily end-of-day (EOD)** series spans 2021-12-20 to 2026-06-21. It carries the prior-day reference price (the basis for band and queue detection), the open, the split-adjusted price, and volume; its length supports a robust historical volatility estimate.

The **options daily EOD** series spans 2024-11-20 to 2026-06-21 (about 380 trading days), comprising 450 contracts (225 calls and 225 puts) across 21 strikes. Strike and maturity are taken directly from the data; Jalali expiry dates are converted to Gregorian for all time-to-maturity calculations.

The **intraday order book** covers the option period. After ingestion it comprises 146 instruments, 13,531 daily CSV files, and 28,033,239 raw rows, from which roughly 12.4 million five-level snapshots were reconstructed, each carrying mid-price, microprice, spread, depth, and order-book imbalance.

The **economic-variables** table supplies the risk-free rate, loaded from the project database (monthly, Jalali calendar, forward-filled to daily). The rate ranges over roughly **30–40% per annum** across the sample, with the latest value about **40%** (Tir 1405 / June 2026); it is used as an annualised continuous-compounding rate throughout.

3.3 Two binding features: illiquidity and a time-varying band

Two structural features dictate the methodology. First, **structured illiquidity**: about **55.8%** of option contract-days within an instrument’s active life have zero volume (56.8% including pre/post-listing rows), concentrated in deep strikes and far maturities. Among traded days, only **50.4%** of (date, strike, maturity) cells have both a call and a put trade on the same day—the precondition for any parity or shadow-price construction—and the fraction of fresh, two-sided intraday snapshots is **49.7%**. Every conclusion is therefore conditional on the liquid sub-universe, and liquidity is a first-class stratification variable rather than a silent filter.

Second, the **price band is time-varying**. Estimated empirically as a high percentile of absolute daily moves on a rolling per-period basis, it is about **4.0%** in the current period (2025–2026) and about **10%** in the earlier period (2022–2023); the change is detected around mid-2025. Queue and band detection therefore use the period-appropriate band rather than a single global value—an important correction, since a fixed band would mislabel queue episodes across the regime change.

Table 1 summarizes the data scale and these gates.

Table 1. Data scale, liquidity, and band.

Category	Metric	Value
Coverage	Total option contracts	450 (225 call, 225 put)
Coverage	Option EOD window	Nov 2024 – Jun 2026 (~380 days)
Coverage	Underlying EOD window	Dec 2021 – Jun 2026
Scale	Order-book rows processed	28,033,239
Scale	Reconstructed 5-level snapshots	~12.4 million
Liquidity	Zero-volume rate (active lifetime)	55.8%
Liquidity	Call-put pair availability (traded days)	50.4%
Liquidity	Intraday-eligible snapshots	49.7%
Band	Current band (2025–2026)	~4.0%
Band	Historical band (2022–2023)	~10%
Rates	Risk-free rate range	~30–40% p.a. (latest ~40%)

4. Methodology

4.1 Analytical pipeline

The analysis is a deterministic twelve-stage pipeline (specs → ingest → book reconstruction → clean → session labelling → as-of sync of each option snapshot to the nearest prior underlying quote within a 120-second window → liquidity flags → band-and-queue classification with episode age τ → pricing → axis_a → axis_b → report). A governing rule is that every analytical result exists simultaneously as a figure, the exact data behind that figure, and a structured findings entry—no number appears only inside a chart.

4.2 Core definitions and parameter choices

Eligibility. A daily-eligible contract-day has strictly positive volume. An intraday-eligible observation is a fresh, two-sided quote within the staleness window.

Moneyness. Moneyness is measured as **K/S** (strike over spot). This is the corrected convention; an earlier S/K definition reversed the apparent direction of the skew and is not used.

Time to maturity. T is computed from the Gregorian expiry under a fixed 365-day count.

Price band and queue detection. The band is estimated empirically on a rolling per-period basis (about 4% currently, about 10% historically). For each option instrument a snapshot is a buy queue when the ask side is empty and the best bid is within one tick of the period ceiling (prior reference $\times (1 + \text{band})$), and symmetrically a sell queue at the floor. A queue must persist for at least three consecutive snapshots, and a five-minute guard window after each release is excluded. Queue age τ is measured in minutes since onset.

Volatility input for the pricing error. To avoid circularity, the BS pricing-error benchmark uses an EWMA volatility ($\lambda = 0.94$) estimated from the five-year underlying return history—independent of the option price being tested.

Risk-free rate. Unlike a naïve $r = 0$ baseline, all BS, parity, and shadow-price calculations use the **real risk-free rate (~30–40% p.a.)** loaded from the economic-variables table. This materially tightens no-arbitrage bounds and shrinks the present value of the strike, and it is central to the parity result below. We flag that the appropriate proxy in an Islamic-finance context (e.g. the Akhza yield versus a bank deposit rate) is itself debatable, and we treat r -sensitivity as a robustness dimension.

Implied volatility. IV is recovered by Brent root-finding and flagged `ok`, `no_arb` (price violates no-arbitrage bounds), `invalid_input` ($T \leq 0$ or $S \leq 0$), or `numerical` (non-convergence).

Shadow price. The parity-implied underlying value is $S^* = C - P + Ke^{-rT}$, defined only where both legs of a (date, strike, maturity) pair trade.

5. Axis A — Black-Scholes Validity in the Free Regime

Axis A tests BS where the underlying is free and continuously tradeable—the fairest test of the model. One caveat sharpens the interpretation: even in free periods the band truncates the underlying’s return distribution, whereas BS assumes an unbounded log-normal. Any residual failure is therefore attributable to band truncation, any genuine volatility structure, and the path-dependent volatility of a daily-rebalanced leveraged ETF.

All Axis A results are conditional on daily-eligible (traded) contract-days, free underlying periods, and the liquidity gate. Of 18,368 eligible observations priced, IV converged (`ok`) for 14,590 (79.4%); 3,321 (18.1%) violated no-arbitrage bounds, 356 (1.9%) had invalid inputs, and 101 (0.6%) failed to converge. The relatively high no-arbitrage-violation share is itself informative: under the real ~30–40% rate the bounds tighten, and many prices that look valid at $r = 0$ violate them once the true discount is applied—evidence that option prices here do not fully embed the high domestic interest rate.

5.1 No statistically significant implied-volatility skew

With moneyness measured correctly as K/S , the slope of near-term (≤ 30 days) call implied volatility against moneyness is -0.179 and is **not statistically significant** (HAC OLS, $t = -1.41$, $p = 0.16$, 95% CI $[-0.427, 0.070]$, $N = 1,744$ contract-days). There is no detectable monotonic skew of the kind routinely reported in developed equity-index markets. This is a correction of independent interest: under the alternative S/K moneyness definition the same data produced a large, highly significant positive slope, which turns out to have been an artifact of the measurement convention rather than a feature of the market.

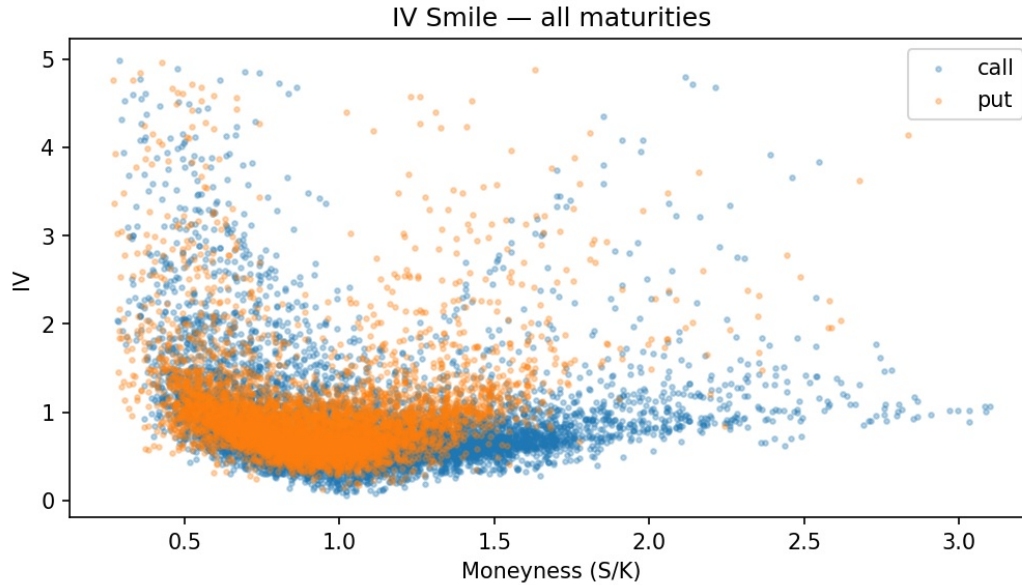


Figure 1. Implied volatility versus moneyness (S/K axis) for all traded contract-days in the free regime, calls (blue) and puts (orange). The cloud is roughly U-shaped with a minimum near the money and wide dispersion at the wings, but no statistically significant monotonic skew once moneyness is measured as K/S .

The surface nonetheless shows real cross-sectional structure. Splitting by maturity (Figure 2), short-dated contracts display a pronounced bowl—elevated IV at both deep out-of-the-money and deep in-the-money wings and a trough near the money—while medium and long maturities flatten. What the data do *not* support is a single monotone trend in moneyness; the dispersion is genuine but not directional. Approximate IV levels by bucket are summarized in Table 2.

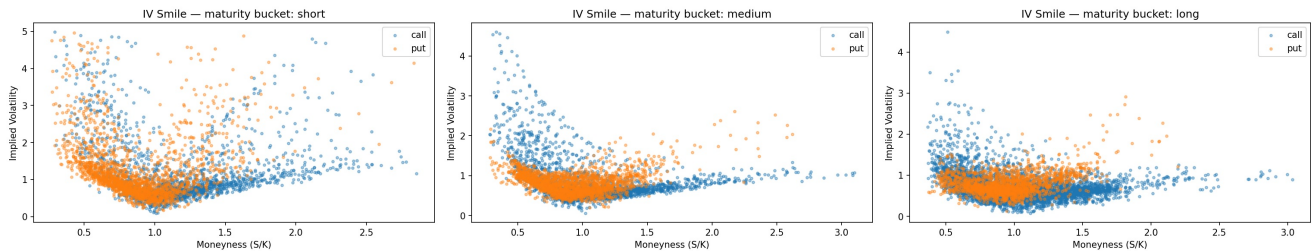


Figure 2. IV smile by maturity bucket — short ($\leq 30d$), medium (30–60d), long ($> 60d$), left to right. The bowl shape is most pronounced for short maturities and flattens with maturity; no consistent monotonic skew emerges.

The maturity dimension is summarized by the IV term structure (Figure 3): mean IV is highest at the shortest expiries—around 2.0–2.5 within a few days of expiry—and falls steeply over the first weeks to a plateau near 0.5–0.8. Calls and puts are broadly intermingled across the term, without one type sitting systematically above the other.

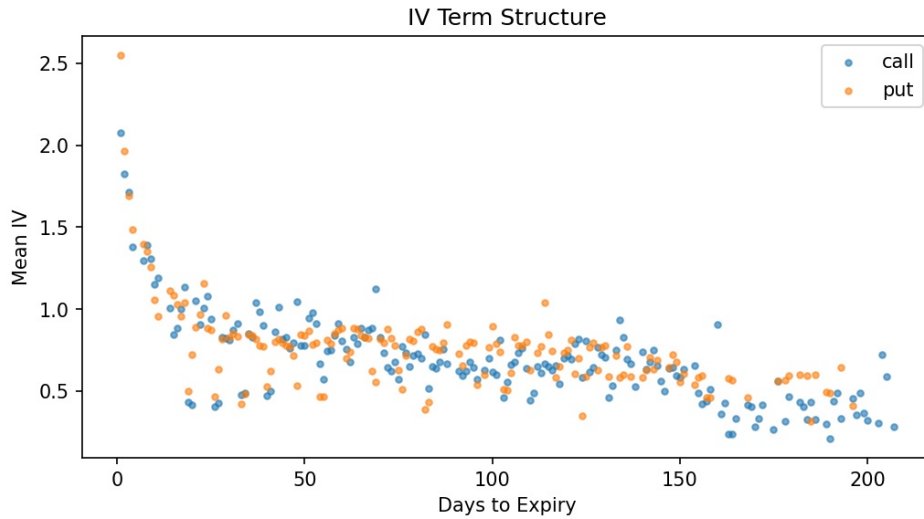


Figure 3. IV term structure: mean implied volatility versus days to expiry, calls (blue) and puts (orange). IV is very high near expiry and decays to a lower plateau; calls and puts overlap across horizons.

Table 2. Approximate implied volatility by moneyness and maturity (read from `axA_iv_summary`; figures indicative).

Moneyness (K/S)	Maturity	Type	Mean IV	Median IV
ATM (0.95–1.05)	short ($\leq 30d$)	call	~0.73	~0.66
ATM	medium	call	~0.78	~0.73
ATM	long	call	~0.80	~0.78
Deep ITM ($K \ll S$)	short	call	high	high
Deep OTM ($K \gg S$)	short	call	elevated	elevated

5.2 Systematic underpricing of both calls and puts

Benchmarked against an independent EWMA volatility and priced with the real risk-free rate, BS **systematically underprices** the options: the median pricing error (market minus model) is **221 rial** and the mean is **452 rial** (HAC t-test: $t = 18.67$, $p = 9.2 \times 10^{-78}$, 95% CI [405, 500], $N = 18,012$). Critically, and unlike the earlier $r = 0$ analysis, the underpricing now spans **both calls and puts** (Figure 4).

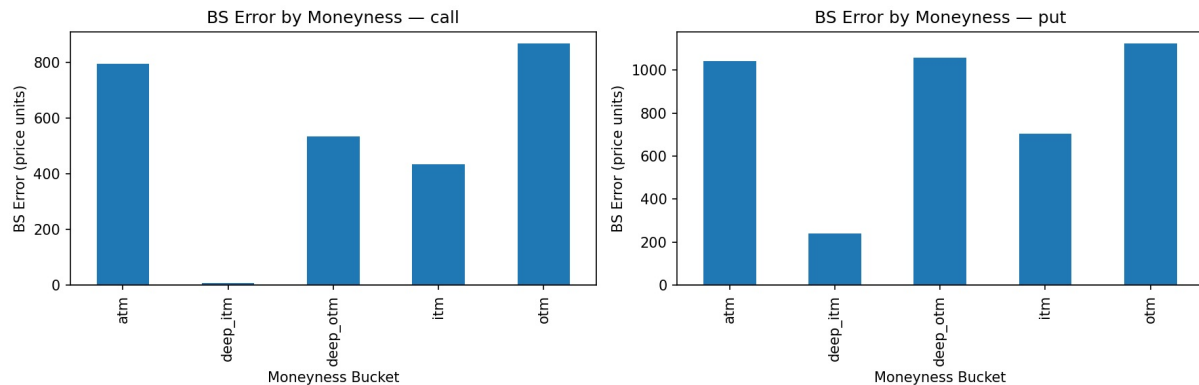


Figure 4. BS pricing error (market – model, EWMA-vol benchmark, real r) by moneyness bucket, calls (left) and puts (right), in price units. Both calls and puts are underpriced across nearly all buckets; deep-ITM is smallest for each type, and out-of-the-money and at-the-money buckets are largest.

For calls, every bucket except deep-ITM is underpriced by several hundred price units, largest out-of-the-money and at-the-money (about 800–870) and near zero for deep-ITM. For puts the underpricing is, if anything, larger—about 1,000–1,100 at the money, out-of-the-money, and deep-out-of-the-money, and smallest (about 235) deep-ITM. That both option types are underpriced points away from a directional, call-specific story and toward a common cause that lifts the fair value of options regardless of type—namely a volatility input that is too low for a leveraged ETF (Table 3).

Table 3. Approximate BS pricing error by moneyness bucket (price units, read from `axA_pricing_error_by_bucket`).

Moneyness bucket	Call error	Put error
ATM	~795	~1,040
OTM	~865	~1,110
Deep-OTM	~530	~1,050
ITM	~430	~705
Deep-ITM	~5	~235

5.3 Negative put-call parity basis under the real rate

Computed with the real risk-free rate, the put-call parity basis, $C - P - (S - Ke^{-rT})$, is significantly **negative**: the median basis is **-332 rial** and the mean is **-515 rial** (HAC t-test: $t = -10.31$, $p = 6.3 \times 10^{-25}$, 95% CI [-613, -417], $N = 6,972$ pairs). The basis distribution (Figure 5) sits at or just left of zero, with a left tail concentrated in the deep-in-the-money and deep-out-of-the-money buckets.

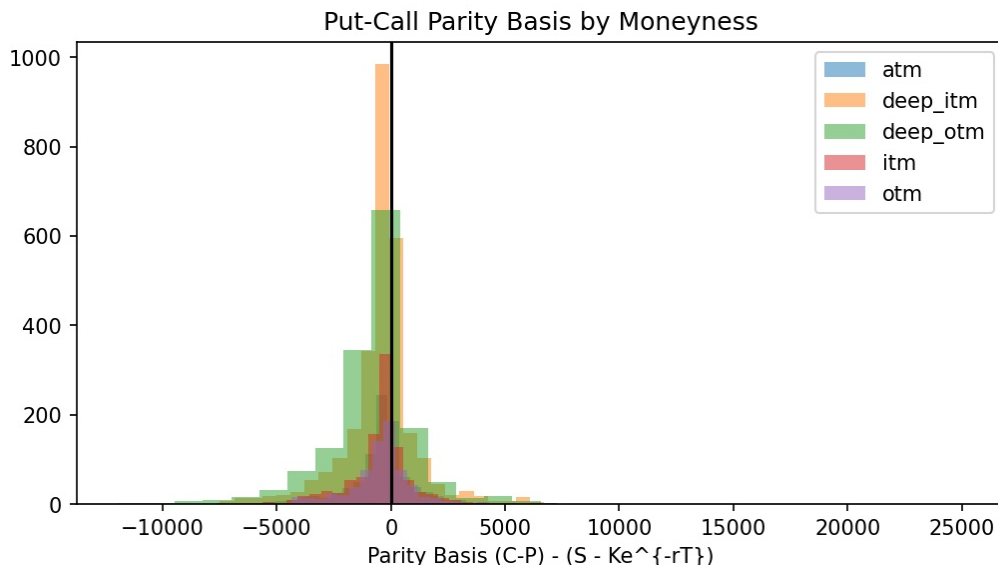


Figure 5. Distribution of the put-call parity basis $C - P - (S - Ke^{-rT})$ by moneyness bucket, using the real r . The mass centers at or just below zero with a left (negative) tail, indicating calls are cheap relative to puts versus what parity requires.

A negative basis means $C - P < S - Ke^{-rT}$: calls are underpriced relative to puts (equivalently, puts are rich relative to calls) versus what parity requires. Two forces make this both possible and persistent. First, the high domestic rate: with r around 30-40%, the present value Ke^{-rT} is markedly below K , so the right-hand side is large and the basis is pushed negative. Second, arbitrage frictions: the trade that would close a negative basis—buy the synthetic forward (long call, short put) and short the underlying—requires short-selling the underlying, which is constrained or prohibited on the TSE. The correcting force is therefore absent, and the wedge survives. This is the opposite sign to the naïve $r = 0$ calculation, which makes the rate treatment, not the option prices alone, the decisive ingredient.

5.4 Cross-contract implied-volatility consistency

A model-adequacy check that needs no external volatility benchmark: under a correct single-volatility model, contracts sharing a date and maturity bucket should imply the *same* volatility, so cross-sectional dispersion should be near zero. It is not (Figure 6). The dispersion distribution concentrates around an IV standard deviation of roughly 0.15-0.35 but carries a long right tail beyond 1.0. The internal consistency is strongly maturity-dependent: long-maturity contracts cluster tightly (a tall mode near 0.15-0.25), whereas short-maturity contracts disperse widely and contribute most of the tail. A single volatility cannot reconcile the strikes that trade on the same day, least of all near expiry—independent evidence against a single-volatility BS description.

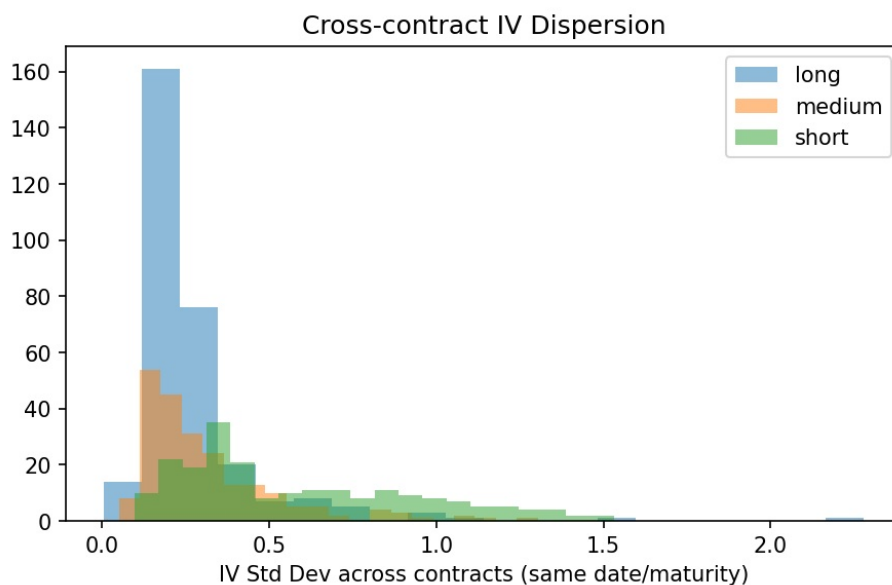


Figure 6. Cross-contract IV dispersion: histogram of the standard deviation of implied volatility across contracts sharing a date and maturity, by maturity bucket. Long-maturity contracts are internally consistent (tight mode near 0.2); short-maturity contracts disperse widely with a heavy right tail.

5.5 Axis A summary

In its most favorable regime, BS is an imperfect model here, but the picture differs from a textbook skew story. There is no statistically significant monotonic IV skew once moneyness is measured correctly; BS underprices both calls and puts; the put-call parity basis is negative under the real rate; and cross-contract IV dispersion is too large for any single volatility to absorb. The first result is a measurement cautionary tale; the second points to a volatility-input bias for a leveraged fund; the third is driven by the high interest rate and arbitrage frictions rather than by option mispricing per se.

6. Axis B — Option Behavior as a Function of Queue Age τ

Axis B examines the regime where the underlying is locked. Over the option window the detector identifies **1,877 queue episodes**, yielding roughly **8,900 queue-regime option snapshots**—about **6,880 buy-queue** and **2,007 sell-queue**—against **7,395,032** free-regime snapshots. Queue states are rare (about 0.12% of all snapshots) but economically central, because they are exactly the moments when the cash market is closed. All Axis B results are conditional on the queue regime, the τ bucket, and moneyness fixed at queue onset; the shadow-price test is further limited by call-put pair availability.

6.1 Depth and order-book imbalance versus τ

By construction a queue is one-sided, so the order-book imbalance (OBI) is pinned at +1.0 in buy queues and −1.0 in sell queues across every τ bucket, and the spread is undefined on the locked side. The informative quantity is depth (Figure 7). In buy queues the median bid-side depth ranges from about 110 units in the youngest buckets to about 155 in the longest (>30 min) episodes, the visible signature of suppressed demand accumulating behind the ceiling; the sell-queue series sits near zero on this bid-depth axis because sell-side pressure shows up on the ask side. The right panel confirms the book is fully one-sided at every age.

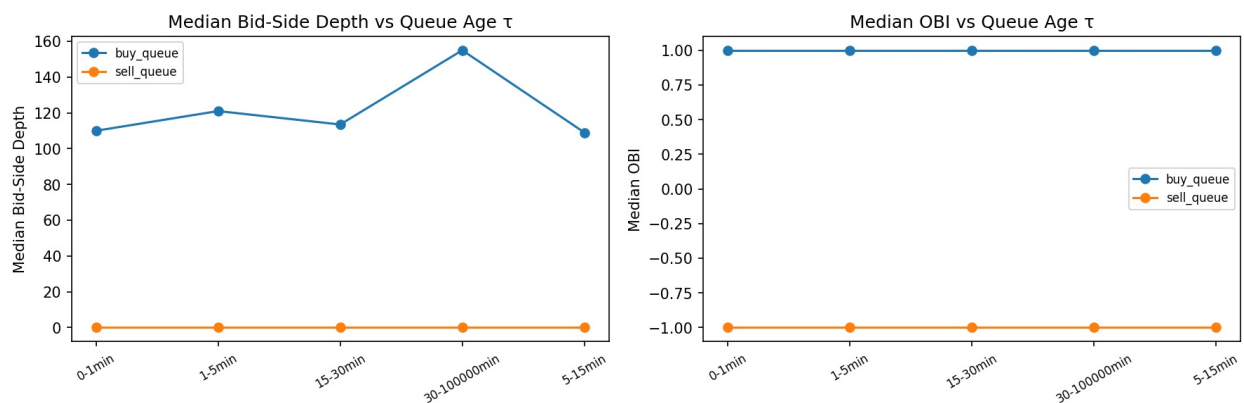


Figure 7. Median locked-side bid depth (left) and median order-book imbalance (right) versus queue age τ , by regime. Buy-queue bid depth ranges ~110–155, highest in the oldest episodes; OBI is pinned at +1 (buy queue) and −1 (sell queue) at every τ .

6.2 Implied volatility compresses with queue age

Joining each queued option's EOD implied volatility to its queue episodes reveals a clear compression of IV with age (Figure 8). Median IV is about **0.94** in the first minute of a queue, falls to about 0.69 in the 1–5 minute bucket and to about **0.34** in the 5–15 and 15–30 minute buckets—roughly a 2.7× decline—before a modest level around 0.45 in the longest episodes.

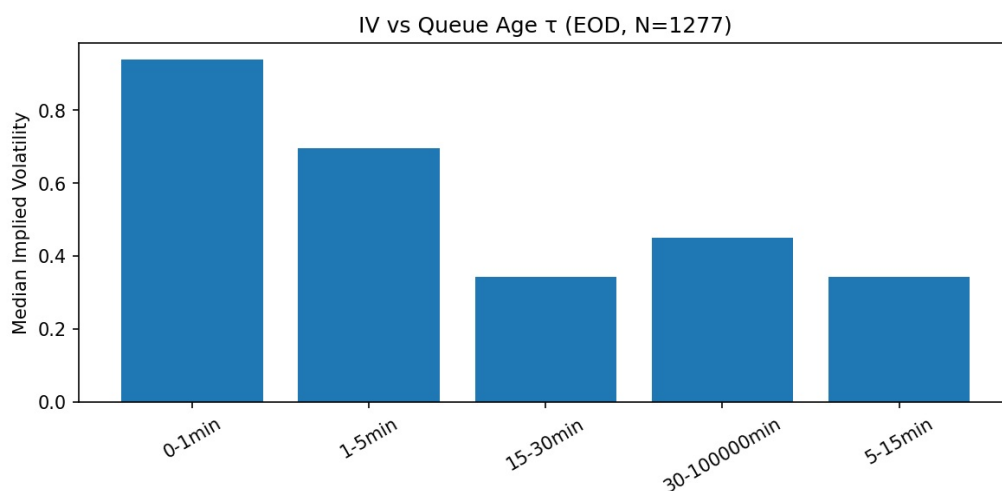


Figure 8. Median end-of-day implied volatility by queue age τ (EOD, $N = 1,277$). IV is far higher in the first minute of a queue (~0.94) than in older buckets (~0.34–0.69).

The natural interpretation is a *certainty effect*. At the instant a queue forms it is uncertain whether the lock will hold or release, and the option prices that ambiguity as high implied volatility. As the lock persists the market becomes increasingly confident it will hold to the close, the option's terminal payoff becomes more predictable, and implied volatility falls. This finding maps EOD IV to

within-day τ , so the τ label reflects the maximum queue age that day rather than a contemporaneous snapshot IV; Section 8 treats this caveat explicitly.

6.3 Quoting activity stays an order of magnitude below the sell side

Order-book update intensity—our proxy for trading activity, since the executed-trade tape was unavailable—tells a complementary story (Figure 9). In buy queues, median activity stays low across all ages, roughly 5–65 updates per day. In sell queues it is about an order of magnitude higher and far more sustained, roughly 465–790 updates per day.

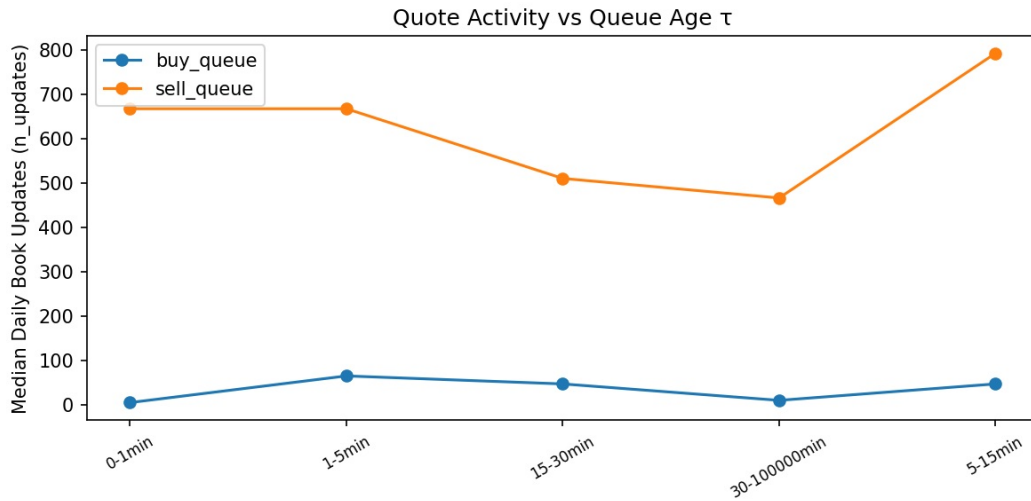


Figure 9. Median daily order-book updates by queue age and regime. Buy-queue quoting (blue) stays low (~5–65/day) at every age, while sell-queue quoting (orange) remains an order of magnitude higher (~465–790/day).

This contradicts a naïve “price discovery migrates to the options” hypothesis for buy-queue locks: when the underlying is locked at the ceiling, its options stay quiet rather than absorbing displaced trading. Sell queues suppress option trading far less—an asymmetry worth flagging for future work.

6.4 The shadow price and next-day-open prediction

The parity-implied shadow price $S^* = C - P + Ke^{-rT}$ is computable for **6,750 EOD pairs** across the queue days. The shadow gap— S^* minus the locked price (the ceiling on a queue day)—is centered near zero with a slight negative tilt and a left tail (Figure 10), consistent with the negative parity basis documented above: with the real rate, the parity-implied value sits close to, and often just below, the locked ceiling rather than well above it.

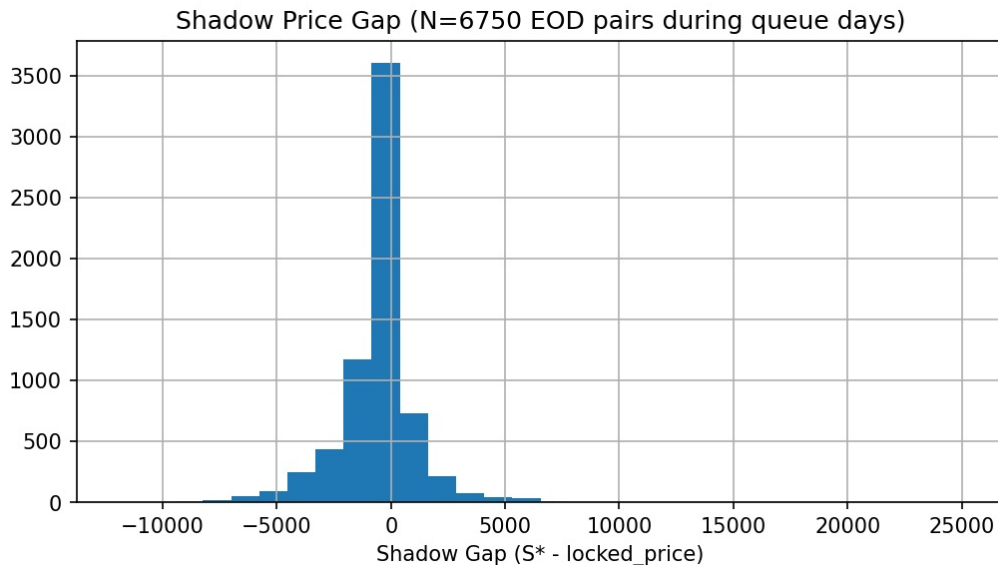


Figure 10. Distribution of the shadow gap $S^* - \text{locked price}$ over 6,750 EOD pairs on queue days, using the real r . The mass is concentrated near zero with a slight negative tilt and a left tail.

The decisive test (the “star test”) regresses the underlying’s next-day open return on the end-of-day shadow gap:

$$\text{next_open_ret} = \alpha + \beta \cdot \text{shadow_gap}.$$

The slope is $\beta = 7.30 \times 10^{-6}$ per rial and is statistically significant ($p \approx 0$, $N = 6,669$). As the scatter (Figure 11) makes plain, the relationship is real but economically negligible: the point cloud is a dense, near-vertical band, so the significance is driven by the large sample rather than by effect size. The honest reading is that the shadow price encodes a faint, statistically detectable signal

about the locked underlying's next-day open, but not one a trader could act on at scale.

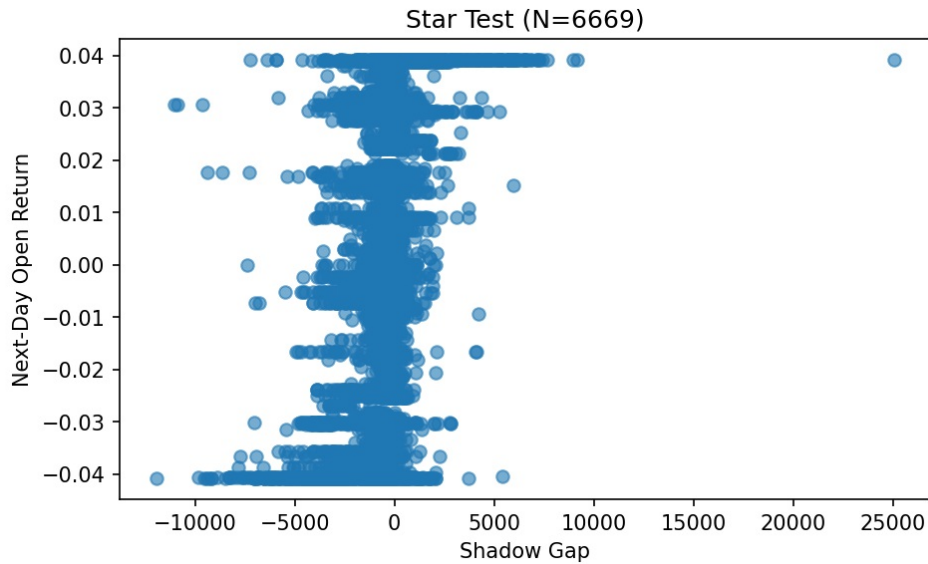


Figure 11. The star test (N = 6,669): next-day open return versus end-of-day shadow gap. A faint positive tilt is present, but the dense vertical band shows the effect is economically negligible despite statistical significance.

Table 4. Axis B headline quantities.

Quantity	Value
Queue episodes	1,877
Buy / sell queue snapshots	~6,880 / ~2,007
Median IV: $\tau < 1$ min $\rightarrow$ 5–30 min	~0.94 $\rightarrow$ ~0.34
Shadow pairs (EOD, queue days)	6,750
Star test — slope β	7.30×10^{-6} per rial
Star test — p / N	$\approx 0 / 6,669$

7. Discussion

The free-regime and queue-regime findings cohere into a single picture, organized around four mechanisms.

Why BS underprices both calls and puts. The pricing error is positive for nearly every bucket of both option types, which rules out a directional, call-specific explanation and points to a common cause that raises the fair value of options regardless of type: a volatility input that is too low. Because Ahrom rebalances daily, its compounded multi-day return is path-dependent and its effective volatility over an option's horizon exceeds a daily-volatility-scaled EWMA estimate (Avellaneda and Zhang, 2010). An EWMA benchmark therefore mechanically *under*-states the volatility the option should be priced on, producing positive errors that are smallest where vega is small (deep-in-the-money, both types) and largest at and out of the money. The error is a feature of the benchmark for a leveraged fund, not necessarily a market inefficiency.

Why the parity basis is negative. This is not a statement about option mispricing per se but about the relationship between calls, puts, and the discounted strike at a high interest rate. With r around 30–40%, Ke^{-rT} is far below K , enlarging $S - Ke^{-rT}$ and pushing the basis negative. Ordinarily arbitrage would close the gap, but the correcting trade requires short-selling the underlying, which is constrained on the TSE, so the wedge persists. The sign of this result is entirely a consequence of using the real rate; at $r = 0$ it reverses. This makes the parity finding a statement about the institutional setting—high rates plus short-selling frictions—rather than about BS itself.

Why implied volatility falls with queue age. The certainty effect explains the compression from about 0.94 in the first minute to about 0.34 thereafter: a fresh queue is informationally ambiguous and priced richly, and as the lock settles the terminal outcome becomes determined and IV declines.

Why activity stays quiet rather than migrating. Buy-queue option quoting stays an order of magnitude below sell-queue quoting at every age, so when the underlying is locked at the ceiling both markets stay quiet together rather than discovery flooding into the options. The shadow price nonetheless encodes a faint, statistically detectable signal about the next-day open—price discovery under a settled lock is better described as a slow leakage of information than a migration of trading.

Taken together, BS is an imperfect descriptive model for this market even in free conditions, but the deviations are best read through the lens of a leveraged-fund volatility process and the high-rate, friction-rich institutional setting, rather than through a textbook skew. The queue regime amplifies, but does not create, these features.

8. Limitations

Several limitations bound the interpretation and are stated plainly.

First, the **risk-free-rate proxy**. The rate (~30–40% p.a.) is a banking-sector series forward-filled to daily frequency. In an Islamic-finance context the appropriate proxy is debatable—the Akhza (أخزا) yield is the conventional candidate, and the database series is a secondary proxy. Because the parity sign and the no-arbitrage-violation share depend directly on r , all rate-sensitive results should be read with an explicit r -sensitivity band.

Second, **no-arbitrage violations**. About 18.1% of priced observations violate no-arbitrage bounds under the real rate (up from roughly 10% at $r = 0$). This suggests either that the effective rate embedded in option prices is lower than the bank rate, or that structural frictions prevent arbitrage, or that prices do not fully incorporate the high rate. We flag it rather than resolve it.

Third, **EOD versus intraday resolution**. Most findings, including IV-versus- τ , map daily EOD quantities to intraday queue episodes by date; the IV compression reflects the day’s IV labelled by the day’s maximum queue age, not a contemporaneous snapshot IV.

Fourth, **selection bias**. Every result is conditional on traded (volume > 0) contract-days; given the 55.8% zero-volume rate, the findings describe the liquid sub-universe.

Fifth, the **band-change date**. The tightening from about 10% to about 4% is detected around mid-2025 by rolling estimation; the precise official date should be confirmed against a TSE announcement.

Sixth, the **European-exercise assumption** for TSE options remains to be confirmed against the rulebook.

Seventh, **leveraged-ETF volatility**. The EWMA benchmark is a downward-biased estimate of the effective volatility for a daily-rebalancing fund; this bias is the leading explanation for the positive pricing error and should be framed as a benchmark feature.

Eighth, the **activity proxy**. Trading intensity uses order-book update counts because the executed-trade tape was unavailable; it measures quoting activity, not realized volume.

The following items remain open: confirmation of option exercise style; the precise band-change date; the appropriate risk-free-rate proxy for the Islamic-finance context; the executed-trade tape; and a genuine intraday IV computation.

9. Conclusion

This study tested the Black–Scholes framework on options written on a daily-rebalancing leveraged ETF in a market governed by hard, time-varying price limits and high interest rates, and tracked those options through the queue episodes the limits produce. In the free regime—the conditions most favorable to the model—BS is imperfect, but the picture is not a textbook skew. Once moneyness is measured correctly, there is no statistically significant monotonic IV skew; BS underprices both calls and puts by a median of 221 rial (mean 452); the put-call parity basis is negative under the real risk-free rate; and cross-contract IV dispersion is too large for any single volatility to absorb. We attribute the underpricing to the leveraged fund’s volatility drag, and the negative parity basis to the high domestic rate combined with short-selling and arbitrage frictions. In the queue regime, implied volatility compresses sharply as a lock ages, consistent with a certainty effect, and buy-queue quoting stays an order of magnitude below sell-queue quoting rather than absorbing displaced price discovery. Yet the parity-implied shadow price retains a statistically significant—if economically negligible—ability to forecast the underlying’s next-day open.

Two broader lessons stand out. Methodologically, a “stylized fact” as basic as the volatility skew can flip sign with the moneyness convention and the risk-free rate, so emerging-market option studies must fix these choices with care. Substantively, a textbook continuous-trading model is the wrong descriptive lens for a market that deliberately interrupts trading, prices a leveraged product, and operates at high interest rates; the queue mechanism amplifies rather than creates the deviations. For practitioners, the shadow price offers a principled—if weak—read on the value locked behind a queue; for researchers, the setting is a clean natural experiment in price discovery under a market lock.

References

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- Black, F., & Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3), 637–654.
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Appendix A. Headline statistics

Table A1. Consolidated results.

Domain	Metric	Value
Band	Empirical band (current / historical)	~4.0% / ~10%
Rates	Risk-free rate	~30–40% p.a. (latest ~40%)
Pricing	Eligible contract-days priced	18,368
Pricing	IV converged (ok)	14,590 (79.4%)
Pricing	no_arb / invalid / numerical	3,321 (18.1%) / 356 (1.9%) / 101 (0.6%)
Pricing	EOD parity pairs	6,972
Axis A	IV skew slope (near-ATM, ≤ 30 d calls, K/S)	-0.179 (p = 0.16, not significant, N = 1,744)
Axis A	BS pricing error — median / mean	221 / 452 rial (p $\approx 10^{-78}$, N = 18,012)
Axis A	Parity basis — median / mean	-332 / -515 rial (p $\approx 10^{-25}$, N = 6,972)
Axis B	Queue episodes	1,877
Axis B	Buy / sell queue snapshots	~6,880 / ~2,007
Axis B	Median IV: $\tau < 1$ min $\rightarrow$ 5–30 min	~0.94 $\rightarrow$ ~0.34
Axis B	Shadow pairs	6,750
Axis B	Star test slope / p / N	7.30×10^{-6} / ≈ 0 / 6,669

Appendix B. Figure inventory

All figures are the pipeline's own output figures (outputs/figures/*.png).

Figure	File	Content
Figure 1	axA_iv_smile	IV versus moneyness, all maturities
Figure 2	axA_iv_smile_short/medium/long	IV smile by maturity bucket
Figure 3	axA_iv_term_structure	Mean IV versus days to expiry
Figure 4	axA_pricing_error_by_bucket	BS pricing error by moneyness, call & put
Figure 5	axA_parity_by_bucket	Parity basis distribution by moneyness
Figure 6	axA_iv_dispersion	Cross-contract IV dispersion by maturity
Figure 7	axB_spread_depth_vs_tau	Locked-side depth and OBI versus τ
Figure 8	axB_iv_vs_tau	Median implied volatility by τ
Figure 9	axB_tradeintensity_vs_tau	Quote-update intensity by τ and regime
Figure 10	axB_shadow_vs_tau	Distribution of the shadow gap
Figure 11	axB_nextday_open	Star test: shadow gap versus next-day open

All numerical results and figures are final pipeline outputs (synced 2026-06-27). The institutional details of the TSE rulebook and the formal r-sensitivity tables remain to be completed by the author.